

Smart Agriculture for Resources Optimization

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INTRODUCTION

The Agriculture is essential for food security, but plant diseases reduce yield and cause losses

Traditional disease detection and pesticide application are manual and time-consuming. This project proposes an AI-based drone system for efficient monitoring. An ESP32-CAM captures crop images and sends them to the cloud for analysis. The system detects diseases, assesses severity, and recommends optimal pesticide dosage..

MOTIVATION

This system detects crop diseases at an early stage using AI and reduces human effort through autonomous drone monitoring. It minimizes pesticide wastage by enabling targeted spraying, thereby saving water and chemicals while promoting precision agriculture.

OBJECTIVE

Traditional plant disease detection relies on manual inspection, which is time-consuming and error-prone. It is not suitable for large-scale agricultural fields. Farmers often lack timely and accurate information about disease type .

METHODOLOGY

Crop images are captured using an ESP32-CAM mounted on a drone. The images are transmitted to the cloud for processing. A CNN-based AI model analyzes the images to detect diseases and assess severity. Based on the results, the system recommends suitable pesticides and optimal dosage

DESIGN

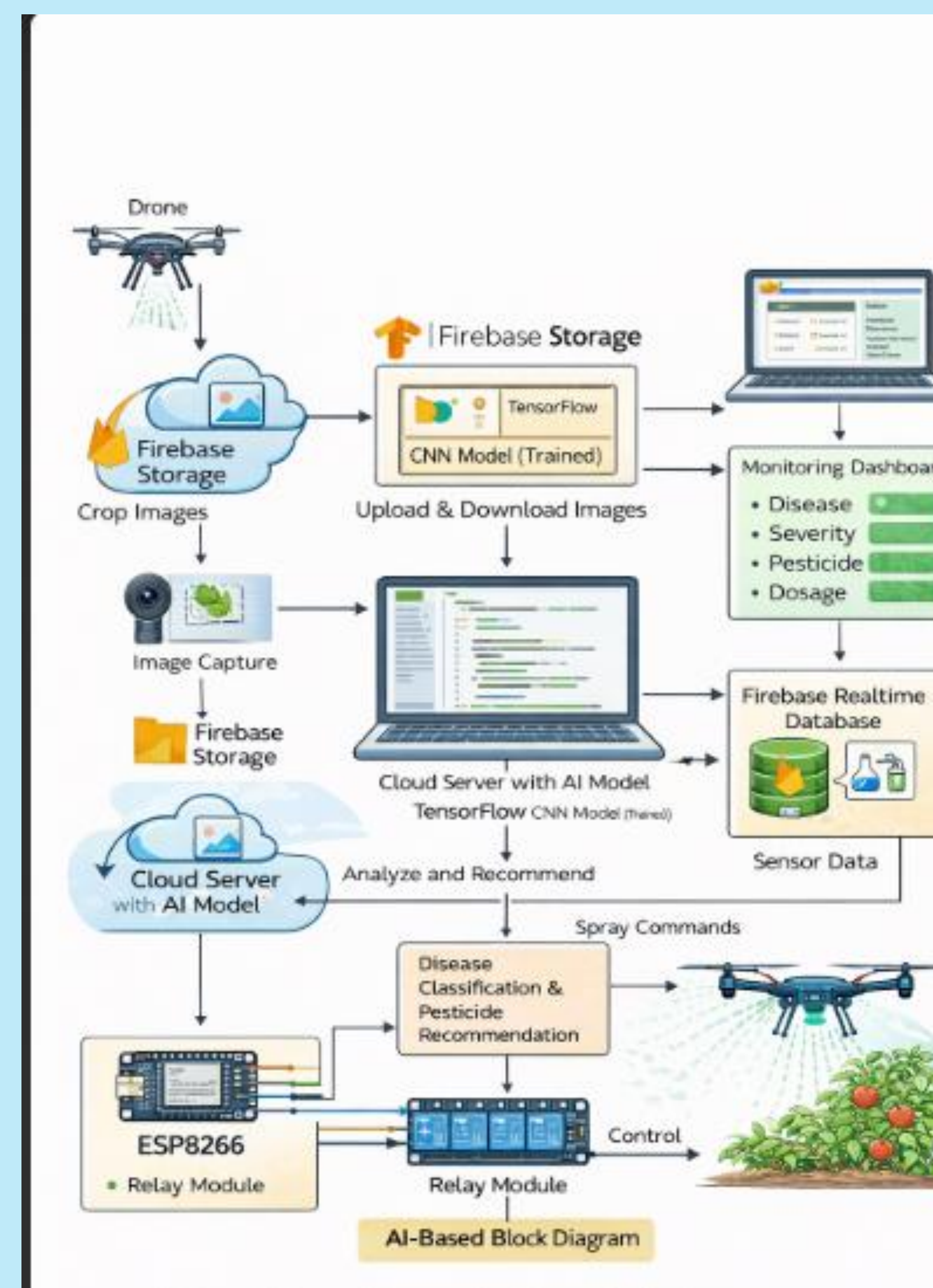


Fig. 1: Architecture Diagram

REALISTIC CONSTRAINTS

The system relies on internet and cloud connectivity for web-based inference, which may be unreliable in rural areas. Pesticide dosage recommendations are based on general guidelines and can vary depending on crop stage, climate, and local farming practices. Additionally, real-time deployment on drones or edge devices is constrained by factors such as battery life, camera quality, and processing capability.

ENGINEERING STANDARDS

1. IEEE 802.11 – Wireless communication for data transmission
2. IEEE 802.15 – Communication standards for UAV and IoT devices
3. ISO 25119 – Safety standards for agricultural electronic systems

RESULTS & INFERENCE

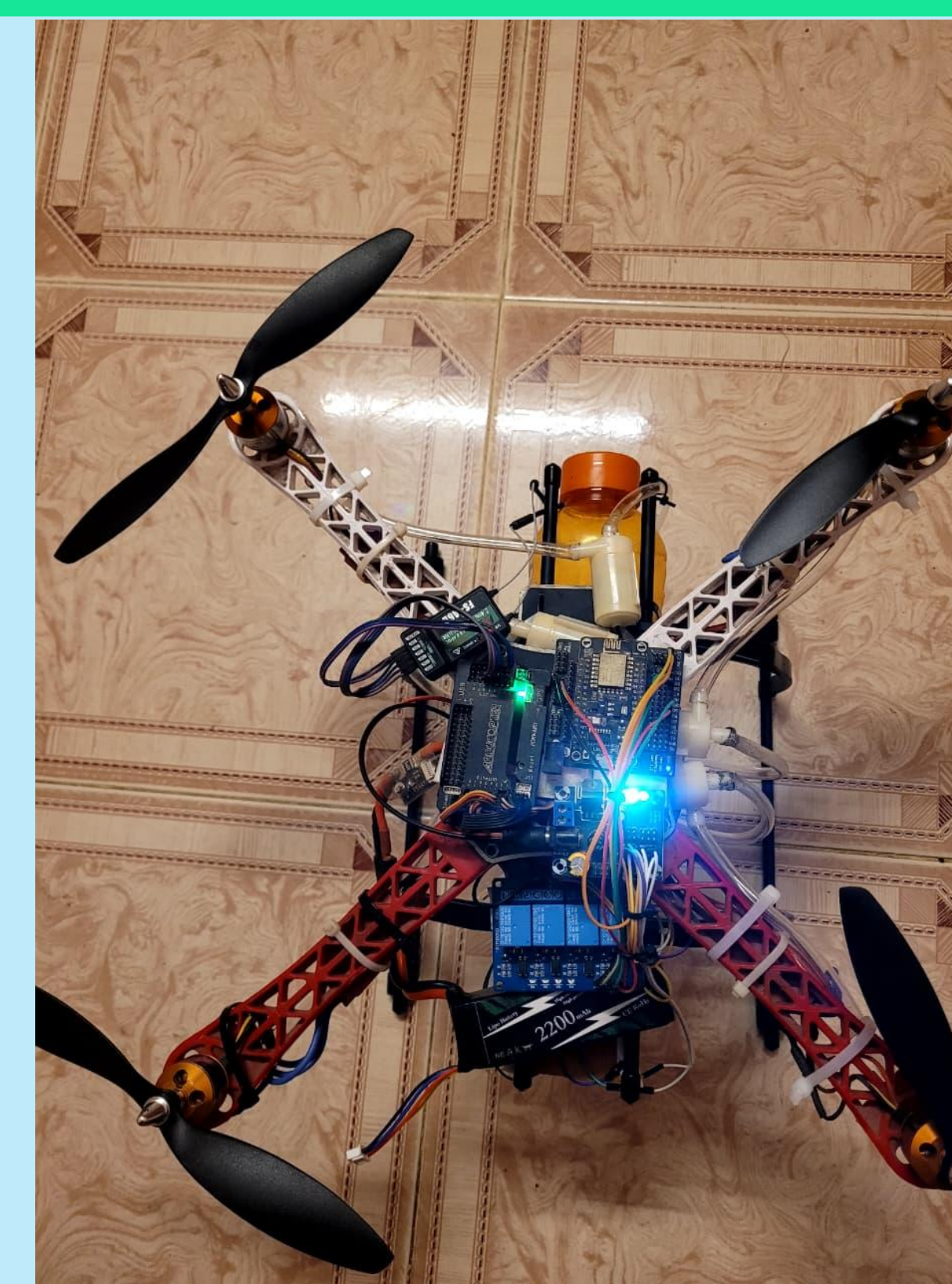


Fig. 2: Hardware Prototype

```
> ... ✓ Firebase Re-initialized Successfully with correct region
✓ Class names loaded successfully: ['Pepper__bell__Bacterial_spot']

🔍 Checking Firebase for new image...
✓ Monitoring completed.
✓ New image downloaded: esp32_images/image_27678.jpg
1/1 _____ 0s 132ms/step

🌿 Disease: Tomato_Early_blight
📊 Severity: 93.11 %
⚠️ Severity Level: Critical
🍷 Pesticide: Chlorothalonil
💧 Dosage: 4 ml per Liter
🗨️ Spray ON
✓ Firebase Updated Successfully
Spray → ON, PWM → 237
Relays → ON, ON, ON, ON
```

Fig 3 : ML Model Output

Table 1. Accuracy, Confidence and Severity Analysis

Accuracy, Confidence and Severity Analysis	Disease, Confidence, Severity, Severity Band			
	Disease Class	Confidence (%)	Severity (%)	Severity Band
Sample 1	Potato Early Bight	99.6	17.17	Medium
Sample 2	Potato Late Bight	43.60	63.55	Very High
Sample 3	Tomato Late Bight	99.03	92.04	Very High
Sample 4	Tomato Early Bight	43.60	17.71	Medium

CONCLUSION

This research presents an intelligent system for accurate plant disease detection and pesticide application using AI and drones. It integrates deep learning, severity estimation, and adaptive dosage recommendation for effective treatment. The system achieves high accuracy (above 95%) with reliable performance in classification and severity assessment.

REFERENCES

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SUSTAINABLE DEVELOPMENT GOALS



CONFERENCE /JOURNAL PUBLICATION

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